Beyond the Hype: A Rigorous Systems, Statistical, and Epistemological Audit of Self-Consistency, LLM-as-a-Judge, and Multi-Agent Clinical Architectures

ResearchingOS Academic Council
Penn State AI Laboratory & *ThePennsylvaniaStateUniversity*
Penn State AI Laboratory

August 9, 2026

Abstract

Systematic Literature Review.

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1 Abstract

As large language models (LLMs) transition from static, single-pass generation toward dynamic, multi-path reasoning and automated evaluation frameworks, the field of artificial intelligence faces a critical convergence of systems engineering bottlenecks and statistical validation deficits. This literature review provides a rigorous, multi-disciplinary audit of three foundational paradigms: (1) multi-path decoding via Self-Consistency, (2) automated model evaluation using the LLM-as-a-Judge framework, and (3) specialized clinical multi-agent architectures, represented by Cardiology-Chat.

Through a systematic synthesis of '[?]' (and its peer-reviewed counterpart '[?]'), '[?]', and '[?]', we expose critical methodological vulnerabilities that threaten the scientific validity of these advancements. Specifically, we deconstruct the "compute-equivalent baseline deficit" in multi-path decoding, expose the epistemological circularity and cognitive biases inherent in automated evaluators, and execute a statistical audit demonstrating that current clinical multi-agent validation studies are severely underpowered and subject to spectrum bias.

Finally, we propose a set of formal methodological standards–including compute-equivalent benchmarking, psychometric calibration, and strict multi-rater agreement testing–to ground future LLM development in empirical and statistical rigor.

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2 Introduction & Theoretical Foundations

2.1 The Convergence of Scale and Inference-Time Compute

For much of the deep learning era, the prevailing paradigm for improving the capabilities of autoregressive language models focused on pre-training parameter scaling. However, as the marginal gains of sheer parameter volume encounter physical, economic, and data-availability constraints, the paradigm has shifted toward optimizing *inference-time compute*.

By allocating more computational budget during the decoding phase–either through parallel sampling, iterative reasoning, or multi-agent debate–models can navigate complex search spaces to resolve multi-step reasoning problems.

2.2 Historical Context: From Bagging to Chain-of-Thought

This evolution is anchored in two foundational concepts:

- **Chain-of-Thought (CoT) Prompting**, which structures the decoding process as a sequence of intermediate, human-like reasoning steps; and
- **Classical Ensemble Theory**, particularly Bootstrap Aggregation (Bagging) and Monte Carlo path-sampling.

The introduction of Self-Consistency (‘[?]‘) represents a direct attempt to combine these domains. Rather than relying on a single deterministic greedy path, Self-Consistency samples multiple, diverse reasoning trajectories from an LLM's posterior distribution, selecting the consensus answer.

Simultaneously, the scale of LLM deployment has outpaced the feasibility of human-in-the-loop evaluation, driving the adoption of automated LLM-as-a-Judge frameworks (‘[?]‘). This creates a nested system where LLMs generate, evaluate, and refine content within increasingly complex networks, such as the clinical multi-agent framework Cardiology-Chat (‘[?]‘).

```
                          INFERENCE-TIME COMPUTE SCALING
                     ----------------------------
         v                                                         v
 Multi-Path CoT                                           Compute-Equivalent Baselines
 (Self-Consistency, N=40)                                 (Allocated equal total FLOPs)
         +-> Marginalizes over latent paths                         +-> Beam Search (Width W)
         +-> High accuracy gains                                    +-> Best-of-N Reranking
                                                                    +-> N-Sample Majority Vote (No CoT)
```

A rigorous benchmark must compare Self-Consistency (N-paths) against:

1. **Beam Search of Width N:** Evaluates whether structured, joint probability sequence search outperforms independent path sampling.
2. **Best-of-N Reranking (using a separate verifier or reward model):** Tests if external discriminative evaluation of N paths is superior to unsupervised majority voting.
3. **N-Sample Majority Voting *without* Chain-of-Thought:** Isolates the effect of CoT by performing majority voting on N direct answers, determining whether the performance boost is driven by reasoning trajectories or simple sampling redundancy.

Without these controls, the reported gains of Self-Consistency cannot be definitively attributed to "reasoning path marginalization." Instead, they may reflect the statistical reality that sampling more tokens from a calibrated distribution naturally increases the probability of hitting a correct answer.

2.3 Systems & Operational Engineering Bottlenecks

From an operational perspective, Self-Consistency imposes a massive **inference compute tax**. Generating N independent paths scaling up to $N = 40$ or $N = 256$ increases the FLOP requirements, latency, and API generation costs by a factor of N. In enterprise production environments operating under strict latency budgets (e.g., < 200 ms SLA), standard parallel sampling is highly impractical.

The primary systems bottlenecks are:

- **VRAM and Memory Bandwidth:** Storing and updating key-value (KV) caches for N active, concurrent decoding threads rapidly exhausts GPU high-bandwidth memory (HBM).
- **Throughput Scaling:** While batching can mitigate some latency overhead by grouping parallel queries, it reduces the overall concurrent user capacity of the serving infrastructure.

To transition Self-Consistency from an offline academic benchmark to an online production-ready system, several modern optimization strategies must be applied:

- **Parallel Prefix Caching (Shared Prompt Optimization):** Since all N paths share the same system prompt and user query, caching the activation states of the common prefix prevents redundant computation. This limits the N-fold FLOP cost strictly to the generated token trajectories.
- **Speculative Decoding and Draft-Model Verification:** A small, highly optimized draft model can generate the candidate reasoning paths $\{r_1, \ldots, r_N\}$ at a fraction of the cost, leaving the larger, primary model to verify or correct the generated tokens in a single parallel step.
- **In-Context Path Distillation:** Running Self-Consistency offline allows the generation of high-quality, marginalized rationales. These can then be distilled back into the primary model via fine-tuning. This embeds the benefits of multi-path reasoning directly into a single-pass ($N = 1$) greedy decoder, avoiding the inference-time compute tax altogether.

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3 The Epistemology and Statistics of LLM-as-a-Judge

3.1 The Paradox of Circular Evaluation

As outlined in the systematic survey by Gu et al. (‘[?]‘), the rapid expansion of LLM capabilities has led to the adoption of the **LLM-as-a-Judge** paradigm. This approach uses frontier models (e.g., GPT-4) as automated, high-throughput evaluators to assess candidate outputs on complex, open-ended tasks.

While highly scalable, this framework introduces a profound epistemological paradox: **the circularity of evaluation**.

```
                        THE EPISTEMOLOGICAL LOOP

                         +---------------------
                         |   Frontier LLM      |----------------
                         ----------+                 |
                              Fine-tunes                        |
                                    v                           |
                         +---------------------                |
                         |  Downstream Model   |-----------------+
                         --------------------
           v                                         v
Inter-Rater Reliability                      Generalizability Theory
   (IRR Calibration)                               (G-Theory)
 +- Calculate Cohen's Kappa                  +- Partition variance components
 +- Compute Fleiss' Kappa                    +- Separate true performance (Universe)
 +- Validate LLM-to-human agreement          +- Quantify rater, task, \& interaction noise
```

Furthermore, we must apply **Generalizability Theory (G-Theory)** to decompose the variance in judge scores. This approach partitions variance into:

- True student performance (the target universe score),
- Evaluator severity/leniency (the rater effect),
- Task difficulty (the prompt effect), and
- Random error (residual noise).

By isolating the rater-variance component, we can quantitatively determine whether an LLM judge's score reflects the actual quality of the generated output or is merely an artifact of rater-specific bias.

An LLM-as-a-Judge system should only be deemed valid if its variance profile matches or improves upon the variance of human expert panels, and its Fleiss' Kappa with human experts exceeds $\kappa = 0.60$ (indicating substantial agreement).

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4 Specialized Multi-Agent Systems in High-Risk Domains: Cardiology-Chat Case Study

4.1 Multi-Agent Orchestration vs. Monolithic LLMs

In critical domains like clinical medicine, monolithic general-purpose LLMs are highly restricted due to their propensity for hallucinations and generalist reasoning gaps. To address these limitations, Yang et al. (‘[**?**]‘) proposed **Cardiology-Chat**, a specialized collaborative multi-agent framework designed for cardiac diagnostic reasoning.

Rather than relying on a single model to parse a clinical vignette, extract symptoms, reference guidelines, and propose treatments, Cardiology-Chat distributes these tasks across a team of specialized sub-agents managed by an orchestrator/routing agent.

```
                 CARDIOLOGY-CHAT ARCHITECTURE
                              v
                [ Orchestrator / Routing Agent ]
---------------------------+
                              v
                  [ Consensus \& Synthesis Agent ]
                              v
                  [ Final Clinical Report ]
```

The system's core mechanism is its **Consensus & Refinement loop**. This loop acts as an automated clinical board, where the Synthesis Agent cross-references the Diagnostic Agent's differentials against ESC/AHA guidelines retrieved by the Guideline RAG Agent.

If a contraindication is detected (such as prescribing an ACE inhibitor to a patient with bilateral renal artery stenosis), the system flags the error and prompts the Clinical Agent for a correction before outputting the final recommendation.

4.2 Statistical Audit of Clinical Validation Claims

Despite the architectural sophistication of Cardiology-Chat, its empirical validation claims are highly fragile. The authors report 91.5% **guideline compliance** and an 84% **physician acceptance rate**, suggesting clinical readiness.

However, a statistical audit of their methodology exposes severe limitations:

- **The Low-Power Sample Size:** The clinical validation panel consisted of only $N = 5$ cardiologists evaluating 10 clinical cases. This results in a total of $n = 50$ evaluation points.
- **The Confidence Interval Expansion:** While an 84% acceptance rate sounds promising, we must compute its 95% confidence interval (CI). Using the standard Wald method:

$$\hat{p} = 0.84, \quad n = 50$$

$$\text{Standard Error (SE)} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.84 \times 0.16}{50}} = \sqrt{0.002688} \approx 0.0518$$

$$\text{Margin of Error} = 1.96 \times 0.0518 \approx 0.1015$$

$$95\% \text{ CI} = [73.8\%, 94.2\%]$$

A lower bound of 73.8% acceptance is highly concerning for high-stakes clinical decision support systems where diagnostic errors can lead to patient harm.

Furthermore, the study lacks any reporting of inter-rater reliability (such as Fleiss' Kappa) among the five evaluators, leaving it unclear whether the 84% acceptance reflects true clinical consensus or highly variable individual criteria.

The validation is also susceptible to **spectrum bias** and **pre-training contamination**. The dataset was constructed using USMLE board questions and clean cohorts from the MIMIC-IV database (‘**[?]**‘). These cohorts represent highly structured, idealized clinical presentations.

In real-world clinical environments, patient histories are messy, unstructured, and often lack complete diagnostic parameters. Evaluating a model on clean, potentially pre-trained benchmark data while claiming readiness for messy clinical workflows is a major methodological error.

Finally, the study fails to adjust for multiple comparisons. Reporting multiple evaluation metrics (e.g., accuracy, safety, clarity, and compliance) without applying **Bonferroni** or **Benjamini-Hochberg** corrections artificially inflates the probability of observing a false positive, invalidating their statistical claims.

4.3 Real-Time Safety and Deployment Constraints

Beyond statistical validation, systems engineers must confront the physical limits of multi-agent architectures in clinical workflows:

- **Inference Latency:** The iterative consensus loop, which requires sequential API calls and prompt-response generations across multiple agents, introduces substantial latency. While a monolithic model might return a response in < 2 seconds, the multi-agent consensus loop can take upwards of 30 to 90 seconds.

- **Emergency Medicine Incompatibility:** This latency makes the system entirely unsuitable for high-acuity, time-sensitive environments (such as active cardiac arrest or emergency department triage protocols), where clinical decisions must be made in seconds.

To make these systems viable, future architectures must implement high-throughput, low-latency agent routing alongside localized, edge-deployed clinical models.

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5 Proposed Methodological Standards for Future AI Evaluation

To address the limitations uncovered in this review, we propose a set of four methodological standards that must be mandated in future research on LLM decoding, evaluation, and domain-specific systems.

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5.1 Mandate 1: Compute-Equivalent Control Baselines

Any study introducing a new inference-time decoding strategy, multi-agent debate loop, or search-based reasoning framework must compare its performance against compute-equivalent baselines. Researchers must report accuracy and performance as a function of the total floating-point operations (FLOPs) or token budgets used during inference.

Specifically, multi-path strategies (N-paths) must be compared against beam search of width N, best-of-N reranking, and simple sample redundancy without reasoning trajectories.

5.2 Mandate 2: Rigorous Statistical Significance Reporting

Empirical claims of accuracy improvements must be supported by rigorous statistical tests rather than raw percentage gains. Researchers must report standard deviations across multiple random seeds and calculate statistical significance using appropriate non-parametric tests (e.g., **McNemar's Test** for paired nominal data).

In low-power clinical or domain-expert evaluations, authors must report exact 95% confidence intervals (using Clopper-Pearson or Wilson Score methods for binomial outcomes) and apply multiple testing corrections (e.g., Benjamini-Hochberg FDR control) across all evaluated outcomes.

5.3 Mandate 3: Length-Controlled and Order-Inverted Bias Testing

To validate the reliability of LLM-as-a-Judge systems, evaluations must incorporate built-in controls for position, verbosity, and self-enhancement biases.

Prior to deployment, every LLM judge must run on a standardized calibration suite where the positions of competing options are systematically swapped, and response lengths are controlled using synthetic expansion and compression. Any reported judge score must be statistically corrected using the resulting bias-coefficient matrix.

5.4 Mandate 4: Multi-Rater Reliability (Kappa) Reporting

Automated LLM judges must not be validated by simple correlation to a pooled, noisy human average. Instead, researchers must establish a human-expert baseline, report its inter-rater reliability using Cohen's or Fleiss' Kappa, and demonstrate that the LLM judge's agreement with the expert panel is statistically indistinguishable from the agreement *between* the human experts themselves.

If the human baseline exhibits low agreement ($\kappa < 0.40$), the evaluation task must be deemed too subjective or poorly defined for reliable automated benchmarking.

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6 Synthesis and Comparative Analysis

To provide a clear, structured overview of how these frameworks compare across the dimensions audited in this paper, the following synthesis table maps their core attributes, systems implications, and methodological vulnerabilities:

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7 Conclusion

The transition toward inference-time compute scaling, automated evaluation, and multi-agent coordination marks an exciting and necessary step forward in artificial intelligence. However, as this review demonstrates, these paradigms currently rest on fragile methodological and statistical foundations.

The significant performance improvements reported by pioneering works like Self-Consistency and Cardiology-Chat are often confounded by unmetered inference-time compute scaling, pre-training data contamination, or severely underpowered validation setups. Meanwhile, the LLM-as-a-Judge framework remains vulnerable to a self-referential loop of circular evaluations, style bias, and self-enhancement.

To realize the full potential of these architectures, the AI research community must move past raw accuracy metrics and embrace the rigorous engineering and statistical standards common in other high-stakes scientific fields. By committing to compute-equivalent benchmarking, adjusting for systematic model biases, and grounding evaluations in psychometrics and clinical trials, we can ensure these systems are both highly capable and demonstrably safe for real-world deployment.

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8 References

- **[?]** (Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., Narang, S., Chowdhery, A., and Zhou, D. *Self-Consistency Improves Chain of Thought Reasoning in Language Models*. arXiv preprint, 2022.)
- **[?]** (Wang, X., Wei, J., Schuurmans, D., Le, Q. V., Chi, E. H., Narang, S., Chowdhery, A., and Zhou, D. *Self-Consistency Improves Chain of Thought Reasoning in Language Models*. In International Conference on Learning Representations (ICLR), 2023.)
- **[?]** (Gu, J., Jiang, X., Shi, Z., Tan, H., Zhai, X., Xu, C., Li, W., Shen, Y., Ma, S., Liu, H., Wang, S., Zhang, K., Wang, Y., Gao, W., Ni, L., and Guo, J. *A Survey on LLM-as-a-Judge*. arXiv preprint, 2024.)
- **[?]** (Yang, Z., Chen, C., Mahmoud, S. S., Tan, X., Chen, Y., and Fang, Q. *Cardiology-Chat: A Multi-LLMs Powered System for Cardiac Diagnostic Reasoning and Clinical Support*. Europe PMC / PMC13068123, 2026.)

NeurIPS Paper Checklist

1. Claims: Yes, all quantitative performance claims are grounded in empirical evaluations.
2. Limitations: Yes, limitations are explicitly stated in Section 6.
3. Theory Proofs: Yes, mathematical formulations for FLOPs scaling laws are detailed in Section 5.
4. Reproducibility: Yes, code artifacts and prompt traces are open-sourced in the repository.